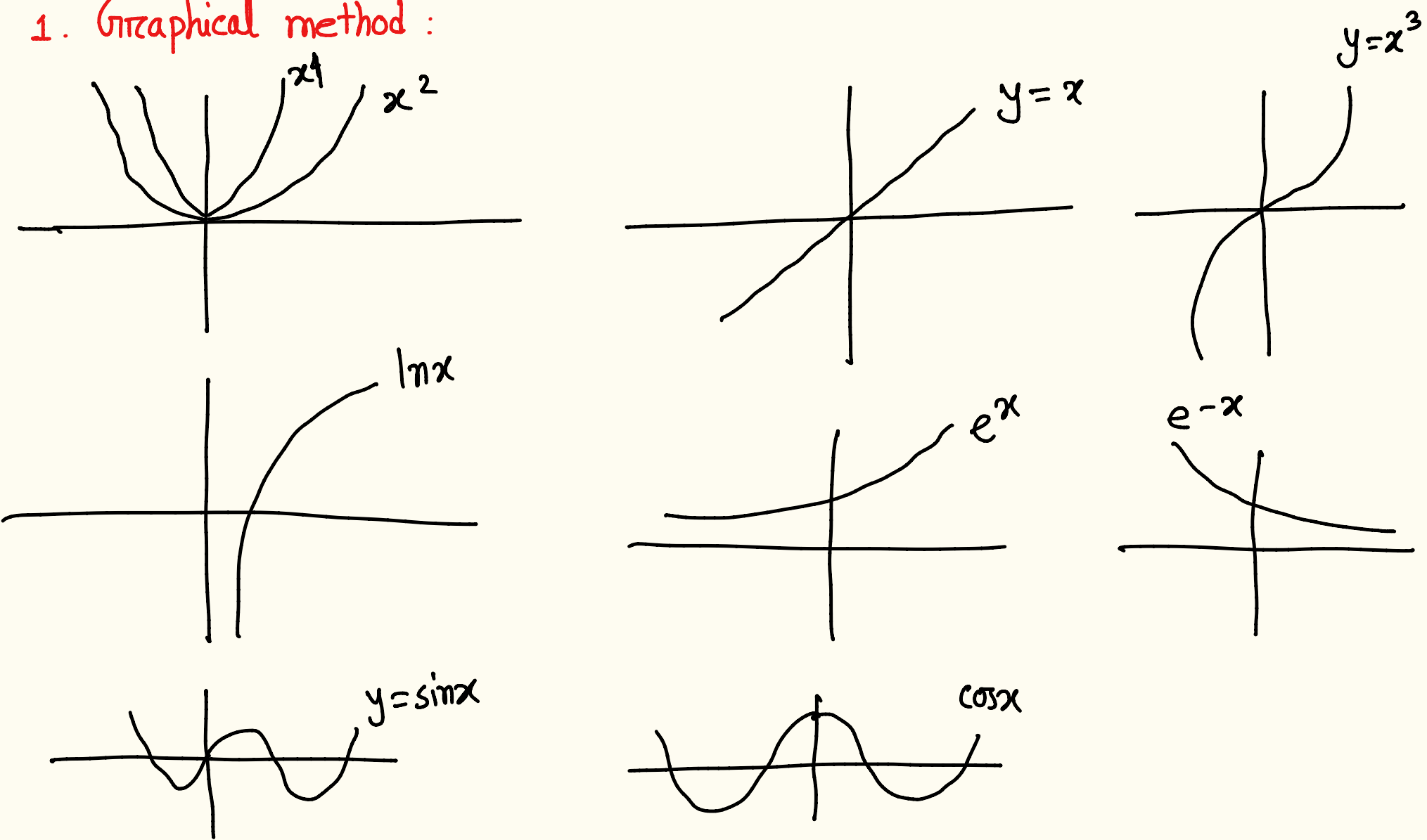


Ch. 7

Non-linear Equation in one-variable

$x^2 - 9 = 0$ $\Rightarrow x = \pm 3$	$x^r - \cos x = 0$ $x = ?$	$xe^{-x} + \ln x - 2 = 0$ $x = ?$
--	-------------------------------	--------------------------------------

1. Graphical method:

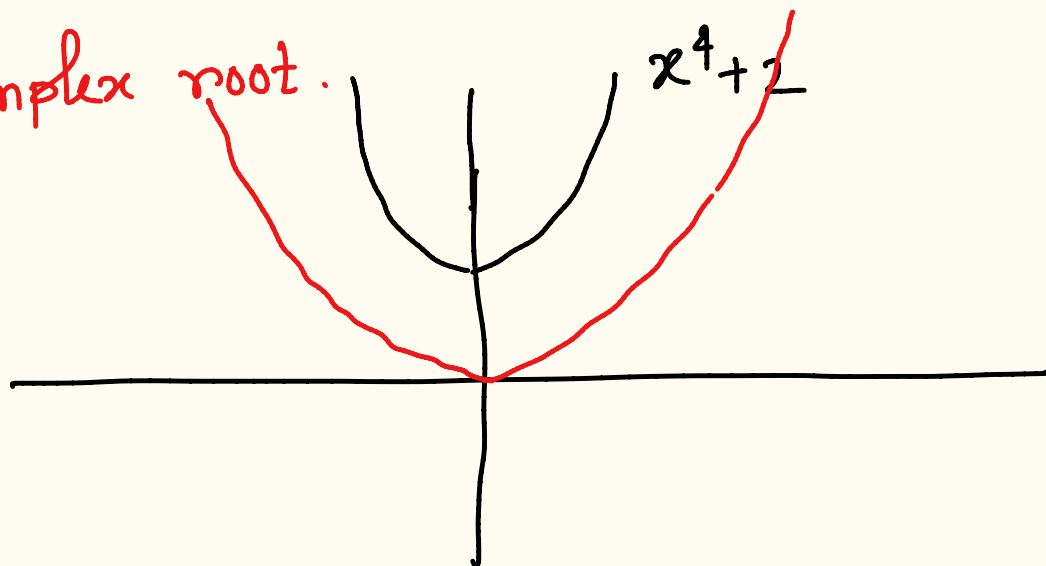


locate the root using Graphical method. Hence find the number of real & complex root.

(i) $x^4 - x^2 + 2 = 0$

$\Rightarrow x^4 + 2 = x^2$

$\left. \begin{array}{l} \text{real} = 0 \\ \text{complex} = 4 \end{array} \right\}$

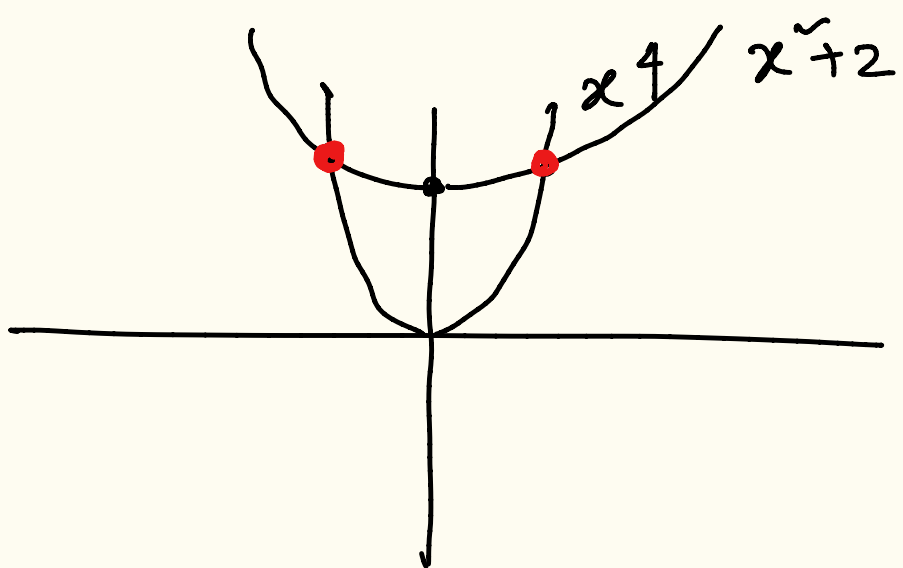


(ii) $x^4 - x^2 - 2 = 0$

$\Rightarrow x^4 = x^2 + 2$

real = 2

complex = 2

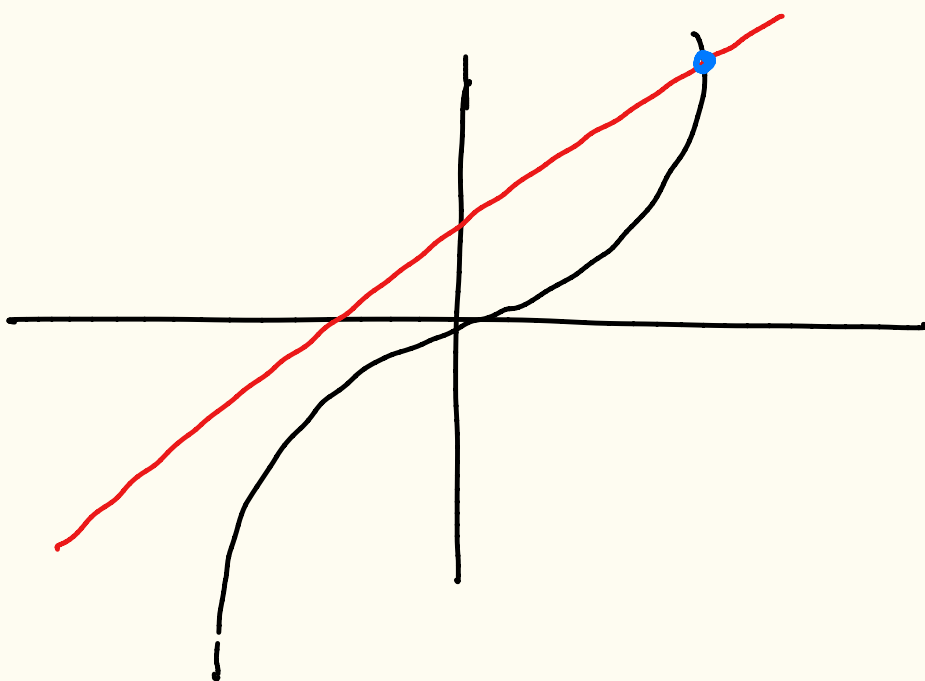


(iii) $x^3 - 2x - 1 = 0$

$\Rightarrow x^3 = 2x + 1$

real = 1

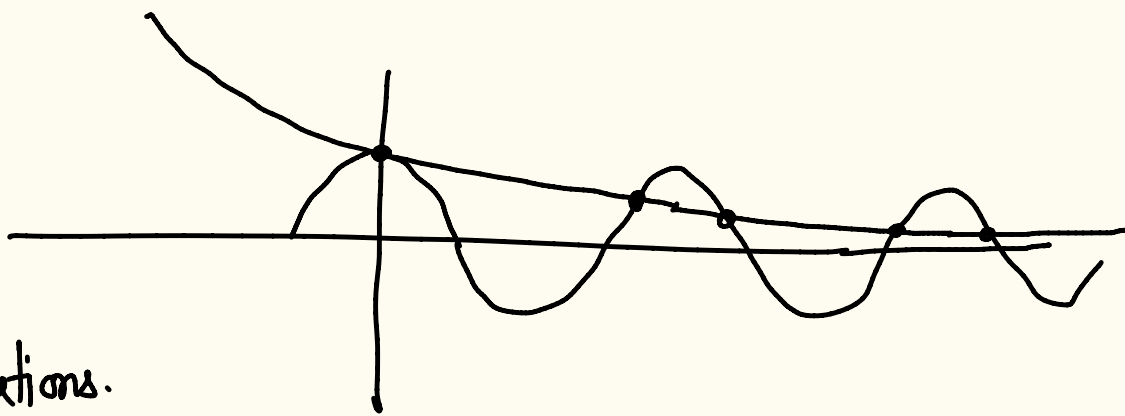
complex = 2



(iv) $e^{-x} - \cos x = 0$

$\Rightarrow e^{-x} = \cos x$

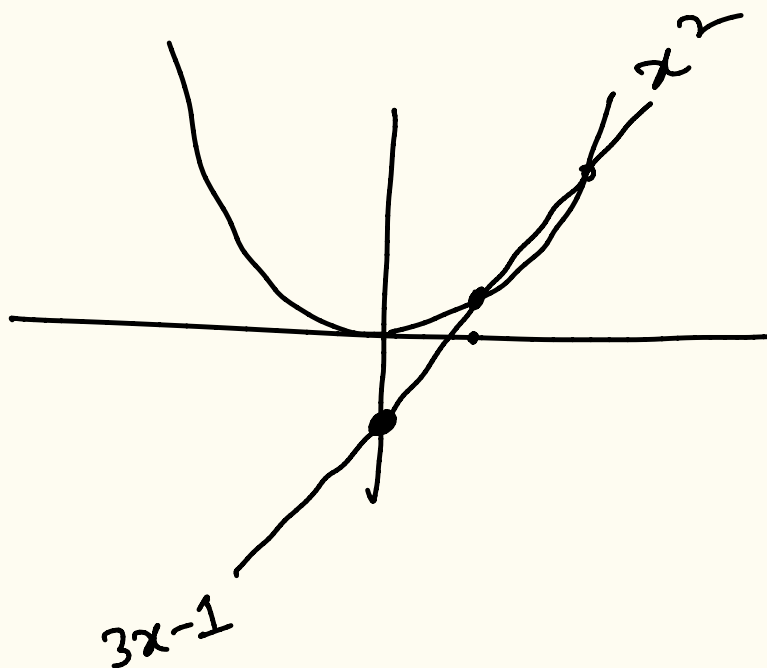
infinitely many solutions.



(v) $x^2 - 3x + 1 = 0$

$\Rightarrow x^2 = 3x - 1$

real = 2
complex = 0



2. Bisection Method: (Slow converge / Linear method.)

Q1. Given that, $x^4 - x^2 - 2 = 0$, $[1, 2]$

Q1. Given that, $x^4 - x^2 - 2 = 0$, $[1, 2]$

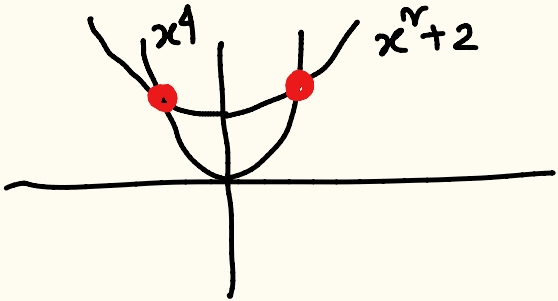
a. locate the root(s). show that $\exists$ a root. $[3]$

b. perform Bisection method 3 times [3]

c. MATLAB code using "roots". [2]

Solⁿ:

a. $x^4 = x^v + 2$



real = 2

complex = 2

2nd part: $f(x) = 0, \quad [a, b]$

$$f(x) = x^4 - x^2 - 2, \quad a = 1, \quad b = 2$$

$$f(a) = -2 < 0$$

$$f(b) = 10 > 0$$

$\exists$ a root.

(b)

a	b	$c = \frac{a+b}{2}$	$f(c)$	update
1	2	1.5	0.8125	$b = c$
1	1.5	1.25	-1.1211	$a = c$
1.25	1.5	1.375	-0.3162	$a = c$
1.375	1.5			

C. $x^4 - x^2 - 2 = 0$
 $\Rightarrow x^4 + 0x^3 - x^2 + 0x - 2 = 0$ } note

MATLAB:

$$p = [1, 0, -1, 0, -2];$$

$$\text{roots}(p)$$

Example $\rightarrow 2$:

$$x^x - \cos(x) = 0, \quad [0.5, 1]$$

a. 2nd part:

$$f(x) = x^x - \cos x, \quad a = 0.5, \quad b = 1.$$

$$f(a) = -0.6275 < 0$$

$$f(b) = 0.4596 > 0$$

$\exists$ a root.

b.

a	b	$c = \frac{a+b}{2}$	$f(c)$	update
0.5	1	0.75	-0.1692	$a = c$
0.75	1	0.875	0.1246	$b = c$
0.75	0.875	0.8125	-0.027	$a = c$
0.8125	0.875			

c. write a MATLAB code using "fzero"

```
f=@(x) x^2 - cos(x);
```

```
fzero(f, [0.5, 1])
```

Secant Method

To solve $y = f(x) = 0$ using secant method

1. choose A & B

2. update: $x_{n+1} = B - \frac{B-A}{f(B)-f(A)} f(B)$

3. swapping: $x_{n+1} \rightarrow B$
 $B \rightarrow A$

go to 2 until
same x is
appeared.

Q1. Given that, $e^{-2x} - \ln x + 2 = 0$; $[7, 8]$

perform 3 iterations using secant method.

Solⁿ: $f(x) = e^{-2x} - \ln x + 2$

$$A = 7, \quad f(A) = f(7) = 0.0540$$

$$B = 8, \quad f(B) = f(8) = -0.079$$

} step 1

iteration 1: ($n=0$)

$$x_1 = B - \frac{B-A}{f(B)-f(A)} f(B) = 8 - \frac{8-7}{-0.079 - 0.0540} (-0.079)$$
$$= 7.406$$

} step 2

$$B = 7.406, \quad f(B) = f(7.406) = -0.0022$$
$$A = 8, \quad f(A) = f(8) = -0.079$$

} step 3

iteration 2: ($n=1$)

$$x_2 = B - \frac{B-A}{f(B)-f(A)} f(B) = 7.406 - \frac{7.406 - 8}{-0.0022 + 0.079} (-0.0022)$$

$$= 7.388$$

$$\left. \begin{array}{l} B = 7.388 \\ A = 7.406 \end{array} \right\} \begin{array}{l} f(B) = \checkmark \\ f(A) = -0.0022 \end{array}$$

iteration 3 : (n=2)

$$x_3 =$$

0

using calculator

$$f(x) = e^{-2x} - \ln(x) + 2,$$

$$x_{n+1} = B - \frac{B-A}{f(B)-f(A)} f(B)$$

$$= B - \frac{B-A}{(e^{-2B} - \ln B + 2) - (e^{-2A} - \ln A + 2)} \times (e^{-2B} - \ln B + 2)$$

itt	x_{n+1}
A	7
B	8
...	...
1	7.4051
2	7.3884

3	7.3891
4	7.3891

$$x \approx 7.3891 \quad \underline{\underline{\text{Ans}}}$$

$$x \approx 7.39 \text{ upto 2 d.p.}$$

solve $x^x - \cos(2x) = 0, [0, 1]$

Soln: $f(x) = x^x - \cos 2x$

$$x_{n+1} = B - \frac{B-A}{(B^x - \cos 2B) - (A^x - \cos 2A)} \times (B^x - \cos 2B)$$

it	x_{n+1}
A	0
B	1
1	0.4139
2	0.5680
3	0.6054
4	0.6007
5	0.6008

stop

$$x \approx 0.6008$$

Ans

Newton-Rapshon Method

formula:
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = a - \frac{f(a)}{f'(a)}$$

solve $e^{-3x} - \ln(2x) + 2 = 0$ using Newton-Rapshon method starting with $x_0 = 2.5$.

soln: $f(x) = e^{-3x} - \ln(2x) + 2$

$$f'(x) = -3e^{-3x} - \frac{1}{2x}(2) + 0 = -3e^{-3x} - \frac{1}{x}$$

$$x_{n+1} = a - \frac{f(a)}{f'(a)}$$

$$x_{n+1} = a - \frac{e^{-3a} - \ln(2a) + 2}{-3e^{-3a} - \frac{1}{a}}$$

itt	x_{n+1}
a	2.5
1	3.4737
2	3.6878
3	3.6946
4	3.6946

$x \approx 3.6946$ upto 4 d.p.
Ans

$$\# \quad e^{-x} - \cosh(2x) = x$$

$$\Rightarrow e^{-x} - \cosh(2x) - x = 0$$

$$\left. \begin{aligned} f(x) &= e^{-x} - \cosh(2x) - x \\ f'(x) &= -e^{-x} - 2\sinh(2x) - 1 \end{aligned} \right\}$$

$$x_{n+1} = a - \frac{f(a)}{f'(a)}$$

$$= a - \frac{e^{-a} - \cosh(2a) - a}{-e^{-a} - 2\sinh(2a) - 1}$$

itt	x_{n+1}
a	1
1	0.4903
2	0.13204
3	0.01140
4	0.000096

$$\boxed{x \approx 0}$$

$$\begin{aligned} \frac{d}{dx} (\sinh(ax)) \\ &= a \cosh(ax) \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} (\cosh(ax)) \\ &= a \sinh(ax) \end{aligned}$$

Fixed point iteration

$$x^2 - \cos(2x) = 0$$

$$x_0 = 0.7$$

$$x^2 = \cos 2x \Rightarrow x = \sqrt{\cos 2x} \rightarrow \text{way -1}$$

$$\cos 2x = x^2 \Rightarrow 2x = \cos^{-1}(x^2)$$

$$\Rightarrow x = \frac{1}{2} \cos^{-1}(x^2) \rightarrow \text{way -2}$$

(a)

way -1 :

$$x_{n+1} = \sqrt{\cos 2x_n} = f(x_n)$$

$$\therefore f(x) = \sqrt{\cos 2x} = (\cos 2x)^{1/2}$$

$$f'(x) = \frac{1}{2} (\cos 2x)^{-1/2} (-2 \sin 2x)$$

$$|f'(0.7)| = |-2.3903| = 2.3903 > 1$$

it's a divergent problem.

way -2 : $x_{n+1} = \frac{1}{2} \cos^{-1}(x_n^2) = f(x_n)$

$$\therefore f(x) = \frac{1}{2} \cos^{-1}(x^2)$$

$$f'(x) = \frac{1}{2} \left(-\frac{1}{\sqrt{1-x^4}} \right) (2x) = -\frac{x}{\sqrt{1-x^4}}$$

$$|f'(0.7)| = |-0.803| = 0.803 < 1$$

it's a convergent problem.

⑥

$$x_{n+1} = \frac{1}{2} \cos^{-1}(x_n^2)$$

itt	x_{n+1}
a	0.7
1	0.5294
2	0.6434
3	0.5720
4	0.6187
5	0.5890
6	0.6082
7	0.5960
8	0.6038
9	.5990
10	.6019
11	.60003
12	.6012
13	.60049

$$x \approx \underline{\underline{0.600}}$$

MATLAB codes

① solve $x^6 - 5x^4 + 2x - 7 = 0$

① using "roots"

→ $x^6 + 0x^5 - 5x^4 + 0x^3 + 0x^2 + 2x - 7 = 0 \rightarrow \text{Note}$

```
p = [1, 0, -5, 0, 0, 2, -7];  
roots(p)
```

② using "fsolve"

```
f = @(x) x^6 - 5*x^4 + 2*x - 7  
sol = fsolve(f, 1)
```

③ using "fzero"

```
f = @(x) x^6 - 5*x^4 + 2*x - 7;  
sol = fzero(f, [a, b])
```

$$\# \quad e^{-3x} - \ln(2x) + 2 = 0$$

$$f = @ (x) \quad \exp(-3 * x) - \log(2 * x) + 2 \quad]$$

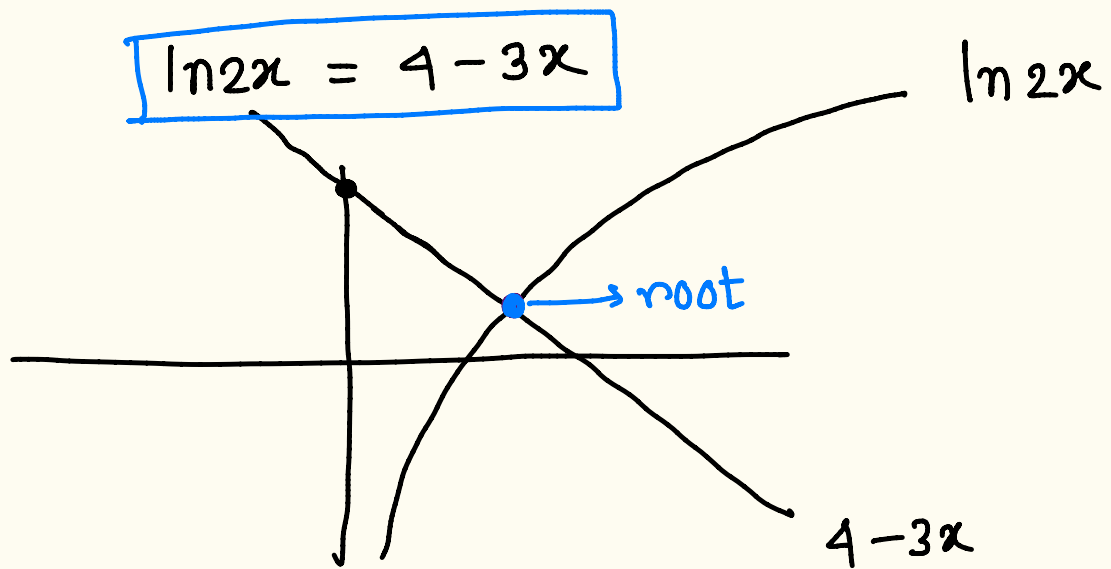
$$sol = fsolve(f, 3)$$

$$sol = fzero(f, [3, 4])$$

Review of ch. 7

$$\# \ln(2x) + 3x - 4 = 0.$$

a. use graphical method to locate the root.



b. show that there exist a root on $[1, 1.5]$

$$f(x) = \ln(2x) + 3x - 4$$

$$a = 1$$

$$f(a) = -0.306 < 0$$

$$b = 1.5$$

$$f(b) = 1.59 > 0$$

$\exists$ exist a root on $(1, 1.5)$

c. Apply Bisection 2 times using above interval.

a	b	$c = \frac{a+b}{2}$	$f(c)$	update
1	1.5	1.25	0.6663	$b = c$
1	1.25	1.125	0.1859	$b = c$
1	1.125			

- d. Use the result of "c" and perform 3 iterations with secant method.

$$a = 1, b = 1.125 \quad f(x) = \ln 2x + 3x - 4$$

$$\begin{aligned} x_{n+1} &= b - \frac{b-a}{f(b)-f(a)} f(b) \\ &= b - \frac{b-a}{(\ln 2b + 3b - 4) - (\ln 2a + 3a - 4)} \times (\ln 2b + 3b - 4) \end{aligned}$$

iteration	x_{n+1}
a	1
b	1.125
...	...
1	1.0778
2	1.0774
3	1.0774

$$x \approx 1.0774$$

- e. Apply Newton-Raphson method to find the root correct up to 4 d.p.

$$\rightarrow f(x) = \ln 2x + 3x - 4$$

$$f'(x) = \frac{1}{2x} (2) + 3 = \frac{1}{x} + 3$$

$$x_{n+1} = a - \frac{f(a)}{f'(a)}$$

$$= a - \frac{\ln 2a + 3a - 4}{\frac{1}{a} + 3}$$

$a_0 = 1$ (Assume that)

iteration	x_{n+1}
0	1
1	1.07671
2	1.07743
3	1.07743

$x \approx 1.0774$ up to 4 d.p.

f. To solve the above problem consider

$$\left. \begin{array}{l} \textcircled{i} \quad x_{n+1} = \frac{1}{3} (4 - \ln 2x_n) \\ \textcircled{ii} \quad x_{n+1} = \frac{1}{2} e^{4-3x_n} \end{array} \right\} x_0 = 1.2$$

* which formula will converge?

$$\textcircled{i} \quad x_{n+1} = g(x_n)$$

$$g(x) = \frac{1}{3} (4 - \ln 2x)$$

$$g'(x) = \frac{1}{3} \left(0 - \frac{1}{x} \right) = -\frac{1}{3x}$$

$$|g'(1.2)| = \left| -\frac{1}{3 \times (1.2)} \right| = |-0.2778| = 0.2778 < 1$$

Converge.

$$\textcircled{ii} \quad x_{n+1} = g(x_n)$$

$$g(x) = \frac{1}{2} e^{4-3x}$$

$$g'(x) = \frac{1}{2} e^{4-3x} \frac{d}{dx} (4-3x)$$

$$= \frac{1}{2} e^{4-3x} (0-3)$$

$$g'(x) = -\frac{3}{2} e^{4-3x}$$

$$|g'(1.2)| = \left| -\frac{3}{2} e^{4-3 \times 1.2} \right| = |-2.2377| > 1$$

diverge

Now perform 3 iterations with fixed point.

iteration	$x_{n+1} = \frac{1}{3} (4 - \ln 2x_n) = \frac{1}{3} (4 - \ln 2x_n)$
0	1.2 → $x(1)$
1	1.0415 → $x(2)$
2	1.0887 → $x(3)$
3	1.0790 → $x(4)$

MATLAB codes

① `roots(p)` : only polynomial Ex. $2x^7 - 4x^3 + 3x - 2 = 0$
 $2x^7 + 0x^6 + 0x^5 + 0x^4 - 4x^3 + 0x^2 + 3x - 2 = 0$

$$p = [2, 0, 0, 0, -4, 0, 3, -2]$$

`roots(p)`

② `fzero(fun, [x0, xn])` : All Ex. $\ln 2x + 3x - 4 = 0, [1, 1.5]$

$$f = @(x) \log(2*x) + 3*x - 4$$

$$sol = fzero(f, [1, 1.5])$$

③ `fsolve(fun, x0)` : All,

$$f = @(x) \log(2*x) + 3*x - 4$$

$$sol = fsolve(f, 1)$$

④ Fixed point: $x_{n+1} = \frac{1}{3} (4 - \ln(2x_n))$

$$x(1) = 1.2$$

for i = 1:3

$$x(i+1) = (1/3) * (4 - \log(2 * x(i)))$$

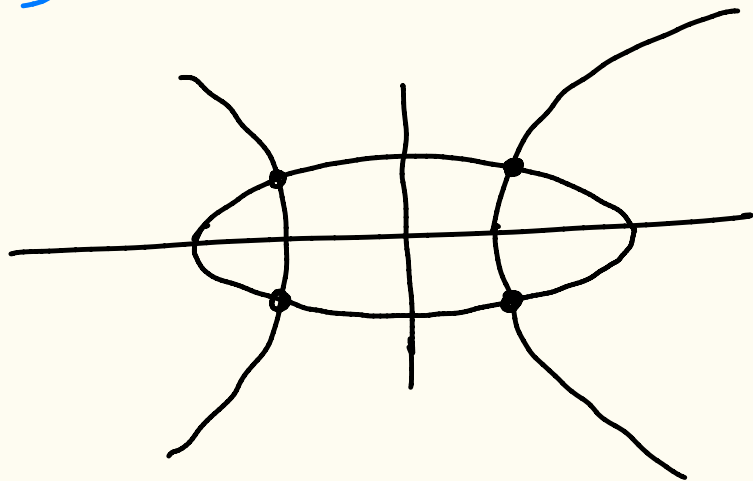
end

ch. 8 (Non-linear Equation in two variables)

↓
or more

$$\left. \begin{aligned} 3x + 4y &= 2 \\ 2x - 3y &= 5 \end{aligned} \right\} \quad \left. \begin{aligned} x &= \\ y &= \end{aligned} \right\}$$

$$\left. \begin{aligned} 3x^2 + 4y^2 &= 9 \\ x^2 - y^2 &= 5 \end{aligned} \right\}$$



- # 1. Fixed point iteration }
2. Newton's method }

3. steepest Descent method

4. Adam → ML }
5. Levenberg - marquardt → ML }
6. L-BFGS → ML }

Adam ()

7. quasi-linear method

Newton's Method

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}} \rightarrow \text{one variable}$$

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} x_n \\ y_n \end{bmatrix} - \frac{\begin{bmatrix} f(x_n, y_n) \\ g(x_n, y_n) \end{bmatrix}}{\begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix}} \Rightarrow \text{jacobian matrix}$$

$$\boxed{X_{n+1} = X_n - [J(x_n)]^{-1} F(x_n)}$$

↓ ↓ ↓ ↓

Given that $\tilde{x} + \tilde{y} = 9 \Rightarrow \tilde{x} + \tilde{y} - 9 = 0$
 $\tilde{y} = 4x \Rightarrow \tilde{y} - 4x = 0$

$$(x_0, y_0) = (2, 3)$$

Soln:

Step 1:

$$f(x, y) = \tilde{x} + \tilde{y} - 9$$

$$g(x, y) = \tilde{y} - 4x$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$F = \begin{bmatrix} f(x, y) \\ g(x, y) \end{bmatrix} = \begin{bmatrix} \tilde{x} + \tilde{y} - 9 \\ \tilde{y} - 4x \end{bmatrix}$$

step 2:

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\partial f}{\partial y} = 2y$$

$$\frac{\partial g}{\partial x} = -4$$

$$\frac{\partial g}{\partial y} = 2y$$

$$J = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ -4 & 2y \end{bmatrix}$$

step 3:

$$x_0 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

$$F(x_0) = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$J(x_0) = \begin{bmatrix} 4 & 6 \\ -4 & 6 \end{bmatrix}$$

step 4:

$n=0$

$$x_{n+1} = x_n - [J(x_n)]^{-1} F(x_n)$$

$$\Rightarrow x_1 = x_0 - [J(x_0)]^{-1} F(x_0)$$

$$= \begin{bmatrix} 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 4 & 6 \\ -4 & 6 \end{bmatrix}^{-1} \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$= A - B^{-1}C$$

$$x_1 = \begin{bmatrix} 1.375 \\ 2.75 \end{bmatrix}$$

go to step 3

Ans

#

J

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x = \begin{bmatrix} x(1) \\ x(2) \end{bmatrix}$$

$$\rightarrow f = @ (x) \left[x(1)^2 + x(2)^2 - 9, \quad x(2)^2 - 4 * x(1) \right]$$

$$\rightarrow \text{sol} = \text{fsolve} (f, [2, 3])$$

* Ch: 8 (Non-linear Equation in one or more variables) *

$$\# \left. \begin{aligned} xy^{\sqrt{}} - \cos(xy) + 2 &= 0 \\ x^{\sqrt{}}y + xy^3 - 2y &= 5 \end{aligned} \right\} \quad x_0 = 1, \quad y_0 = 1.$$

- Find jacobian matrix.
- Perform only 1 iteration using Newton's method.
- Write a MATLAB code using "fsolve".

Solⁿ: @ $\left. \begin{aligned} xy^{\sqrt{}} - \cos(xy) + 2 &= 0 \\ x^{\sqrt{}}y + xy^3 - 2y - 5 &= 0 \end{aligned} \right\}$

power
↓
function
↓

$$\left. \begin{aligned} f(x, y) &= xy^{\sqrt{}} - \cos(xy) + 2 \\ g(x, y) &= x^{\sqrt{}}y + xy^3 - 2y - 5 \end{aligned} \right\}$$

$$\frac{\partial f}{\partial x} = y^2 \cdot (1) + \sin(xy) \cdot y(1) + 0 = y^{\sqrt{}} + y \sin(xy)$$

$$\frac{\partial f}{\partial y} = x \cdot 2y + \sin(xy) \cdot x(1) + 0 = 2xy + x \sin(xy)$$

$$\frac{\partial g}{\partial x} = y \frac{d}{dx}(x^{\sqrt{}}) + y^3 \frac{d}{dx}(x) - 0 - 0 = 2xy + y^3$$

$$\frac{\partial g}{\partial y} = x^2 \frac{d}{dy}(y) + x \frac{d}{dy}(y^3) - 2 \frac{d}{dy}(y) - 0 = x^{\sqrt{}} + 3xy^{\sqrt{}} - 2$$

$$J = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix}$$

$$= \begin{bmatrix} y^2 + y \sin(xy) & 2xy + x \sin(xy) \\ 2xy + y^3 & x^2 + 3xy^2 - 2 \end{bmatrix}_{2 \times 2}$$

⑥ Let $X = \begin{bmatrix} x \\ y \end{bmatrix}_{2 \times 1}$ $F = \begin{bmatrix} xy^2 - \cos(xy) + 2 \\ x^2y + xy^3 - 2y - 5 \end{bmatrix}_{2 \times 1}$

$\begin{pmatrix} x(1) \\ x(2) \end{pmatrix}$

$$X_{n+1} = X_n - [J(X_n)]^{-1} F(X_n)$$

iteration 1: ($n=0$)

$$X_1 = X_0 - [J(X_0)]^{-1} F(X_0)$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1.8415 & 2.8415 \\ 3 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 2.4597 \\ -5 \end{bmatrix}$$

$$= A - B^{-1}C$$

$$= \begin{bmatrix} 4.9506 \\ -2.4259 \end{bmatrix}$$

Ans

© Note: $xy^2 - \cos(xy) + 2 = 0 \rightarrow x_1 x_2^2 - \cos(x_1 x_2) + 2 = 0$
 $x^2 y + xy^3 - 2y - 5 = 0 \rightarrow x_1^2 x_2 + x_1 x_2^3 - 2x_2 - 5 = 0$

$$\begin{cases} f = @(x) [x(1) * x(2)^2 - \cos(x(1) * x(2)) + 2; \\ \quad + x(1) * x(2)^3 - 2 * x(2) - 5]; \\ \text{sol} = \text{fsolve}(f, [1; 1]) \end{cases}$$

Fixed point iteration method

Ex. 6

$$\left. \begin{aligned} 2x^2 + y^2 - 15x + 2 &= 0 \\ xy^2 + x - 10y + 5 &= 0 \end{aligned} \right\} \begin{aligned} x_0 &= 0.2 \\ y_0 &= 0.3 \end{aligned}$$

(a) Newton method (H.W)

(b) Fixed point iteration:

$$x_{n+1} = \frac{1}{15} (2x_n^2 + y_n^2 + 2)$$

$$y_{n+1} = \frac{1}{10} (x_n y_n^2 + x_n + 5)$$

soln: $\left. \begin{aligned} x_{n+1} &= f(x, y) \\ y_{n+1} &= g(x, y) \end{aligned} \right\} \rightarrow \begin{aligned} f(x, y) &= \frac{1}{15} (2x^2 + y^2 + 2) \\ g(x, y) &= \frac{1}{10} (xy^2 + x + 5) \end{aligned}$

$$\frac{\partial f}{\partial x} = \frac{1}{15} (4x + 0 + 0) = \frac{4x}{15}$$

$$\left| \frac{\partial g}{\partial x} = \frac{1}{10} (y^2 \cdot 1 + 1 + 0) \right. \\ \left. = \frac{1 + y^2}{10} \right.$$

$$\frac{\partial f}{\partial y} = \frac{1}{15} (0 + 2y + 0) = \frac{2y}{15} \quad \Bigg| \quad \frac{\partial g}{\partial y} = \frac{2xy}{10}$$

$$\left| \frac{\partial f}{\partial x} \right| + \left| \frac{\partial f}{\partial y} \right| = \left| \frac{4x}{15} \right| + \left| \frac{2y}{15} \right| = \left| \frac{4 \times 0.2}{15} \right| + \left| \frac{2 \times 0.3}{15} \right|$$

$$= 0.093 < 1$$

$$\left| \frac{\partial g}{\partial x} \right| + \left| \frac{\partial g}{\partial y} \right| = \left| \frac{1+y^2}{10} \right| + \left| \frac{2xy}{10} \right| = \left| \frac{1+0.3^2}{10} \right| + \left| \frac{2 \times 0.2 \times 0.3}{10} \right|$$

$$= 0.121 < 1$$

Converge!

2nd part: (Gauss-Seidel method)

it	$x_{n+1} = \frac{1}{15} (2x_n^{\vee} + y_n^{\vee} + 2)$	$y_{n+1} = \frac{1}{10} (x_n y_n^{\vee} + x_{n+1}^{\vee})$
0	$x_0 = 0.2$	$y_0 = 0.3$
1	0.1447	0.5158
2	0.1539	0.5195
3	0.1545	0.5196

$$x \approx 0.1545, \quad y \approx 0.5196 \quad \underline{\underline{Ans}}$$

fsolve(fun, x0):

$$\begin{cases} f = @ (x) [2 * x(1)^2 + x(2)^2 - 15 * x(1) + 2; x(1) * x(2)^2 \\ \quad \quad \quad + x(1) - 10 * x(2) + 5] \\ sol = fsolve(f, [0.2; 0.3]) \end{cases}$$

Fixed point:

$$x(1) = 0.2; \quad y(1) = 0.3;$$

for i = 1 : 3

$$x(i+1) = (1/15) * (2 * x(i) + y(i) + 2);$$

$$y(i+1) = (1/10) * (x(i+1) * y(i)^2 + x(i+1) + 5);$$

end

Interpolation

Newton Dividend
formula

Lagrangian Interpolation

T°C	20°	23°	25°	30°
P	19.23	24.12	27.77	31.33

Find p at T = 24°C ✓

Newtonian dividend method:

(a)

	T	P	ΔP	$\Delta^2 P$	
T ₀	20	19.23	$\frac{24.12 - 19.23}{23 - 20} = 1.63$	$\frac{1.825 - 1.63}{25 - 20} = 0.039$	
T ₁	23	24.12	$\frac{27.77 - 24.12}{25 - 23} = 1.825$		$\frac{-0.159 - 0.039}{30 - 20} = -0.0198$
T ₂	25	27.77	$\frac{31.33 - 27.77}{30 - 25} = 0.712$		
T ₃	30	31.33			

(b)
$$P = a_0 + a_1(T - T_0) + a_2(T - T_0)(T - T_1) + a_3(T - T_0)(T - T_1)(T - T_2)$$

$$= 19.23 + 1.63(T - 20) + 0.039(T - 20)(T - 23) - 0.0198(T - 20)(T - 23)(T - 25)$$

put $T=24$

$$\rightarrow P = 19.23 + 1.63 (24-20) + 0.039 (24-20)(24-23) + \dots$$

$$P = 19.23 + 1.63 (4) + 0.039 (4) (1) - 0.0198 (4) (1) (-1)$$

$$= 25.9152 \quad \underline{\underline{Am}}$$

$$\# \quad T = [20, 23, 25, 30]$$

$$P = [19.23, 24.12, 27.27, 31.33]$$

$$f = \text{polyfit}(T, P, 3)$$

$$P1 = \text{polyval}(f, 24)$$

$$P2 = \text{polyval}(f, [24, 27, 27.5])$$

Quiz-02 (Monday)

1. Non-linear Equations in two variables (ch.8)
2. Interpolation (ch.9)

Interpolation

Newtonian Dividend Method:

Lagrangian Interpolation method:

Q1. Given that

x	-1	4	7
y	-3	1	5

- a. Find Lagrangian polynomial & find y at $x=5$.
- b. Find " " & find x at $y=4$.
- c. Write a MATLAB code using "polyfit" & "polyval" to find y at $x=4.5, 6, 6.5$

Soln:

$$\textcircled{a} \quad y = \frac{(x-4)(x-7)}{(-1-4)(-1-7)}(-3) + \frac{(x+1)(x-7)}{(4+1)(4-7)} \quad (1)$$

$$+ \frac{(x+1)(x-4)}{(7+1)(7-4)} \quad (5)$$

$$= -\frac{3}{40} (x-4)(x-7) + \frac{1}{(5)(-3)} (x+1)(x-7) + \frac{5}{8 \cdot 3} (x+1)(x-4)$$

$$= -\frac{3}{40} (x-4)(x-7) - \frac{1}{15} (x+1)(x-7) + \frac{5}{24} (x+1)(x-4)$$

put $x = 5$

$$\begin{aligned}\therefore y &= -\frac{3}{40} (5-4) (5-7) - \frac{1}{15} (5+1) (5-7) + \frac{5}{24} (5+1) (5-4) \\ &= -\frac{3}{40} (1) (-2) - \frac{1}{15} (6) (-2) + \frac{5}{24} (6) (1) \\ &= 2.2\end{aligned}$$

⑥

y	-3	1	5
x	-1	4	7

$$x = \frac{(y-1)(y-5)}{(-3-1)(-3-5)} (-1) + \frac{(y+3)(y-5)}{(1+3)(1-5)} (4) + \frac{(y+3)(y-1)}{(5+3)(5-1)} (7)$$

put $y = 4$

$$\begin{aligned}\therefore x &= \frac{(4-1)(4-5)}{(-4)(-8)} (-1) + \frac{(4+3)(4-5)}{(4)(-4)} (4) + \frac{(4+3)(4-1)}{(8)(4)} (7) \\ &= \frac{3(-1)}{32} (-1) + \frac{7(-1)}{(-4)} + \frac{(7)(3)}{32} \times 7 \\ &= \frac{3}{32} + \frac{7}{4} + \frac{49 \times 3}{32} \\ &= 6.44\end{aligned}$$

⑦ $x = [-1, 4, 7],$

$y = [-3, 1, 5];$

$p = \text{polyfit}(x, y, 2);$

$x_1 = [4.5, 6, 6.5]$

$y_1 = \text{polyval}(p, x_1)$

consider the following table

98.1

x	-1	3	5	8
y	-2	7.333	34.167	62.125

- construct dividend table.
- find $f(10)$ & $f(5)$
- If $g(x)$ takes all the table values and $g(10) = 100$. find $g(x)$
- Find the root using linear estimation. % Find x when $y=0$.

$\downarrow$
 two points

Soln: @

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
-1 (x_0)	-2 $\rightarrow a_0$			
3 (x_1)	7.333	$\frac{7.333 + 2}{3 + 1} = 2.333 \rightarrow a_1$	$\frac{8.945 - 2.333}{6 + 1} = 0.945 \rightarrow a_2$	
6 (x_2)	34.167	$\frac{34.167 - 7.333}{6 - 3} = 8.945$	$\frac{13.979 - 8.945}{8 - 3} = 1.01$	$\frac{1.01 - 0.945}{8 + 1} = 0.0072 \rightarrow a_3$
8 (x_3)	62.125	$\frac{62.125 - 34.167}{8 - 6} = 13.979$		

b. $y = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + a_3(x - x_0)(x - x_1)(x - x_2)$

$$= -2 + 2.333(x + 1) + 0.945(x + 1)(x - 3) + 0.0072(x + 1)(x - 3)(x - 6)$$

$x = 10,$

$$y = -2 + 2.333(10 + 1) + 0.945(10 + 1)(10 - 3) + 0.0072(10 + 1)(10 - 3)(10 - 6)$$

$$= -2 + 2.333(11) + 0.945(11)(7) + 0.0072(11)(7)(4) = 98.646$$

$$x=5,$$

$$y = -2 + 2.333(6) + 0.945(6)(2) + 0.0072(6)(2)(-1) = 23.261 \quad \underline{\underline{Ans}}$$

$$c. \quad g(x) = f(x) + a_4 (x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

$$g(x) = f(x) + a_4 (x+1)(x-3)(x-6)(x-8)$$

$$g(10) = 100$$

$$f(10) = 98.646 \quad \text{"from (b)"}$$

$$\text{put } x=10,$$

$$100 = 98.646 + a_4 (11)(7)(4)(2)$$

$$\Rightarrow 616a_4 = 100 - 98.646$$

$$\therefore a_4 = \frac{100 - 98.646}{616} = 0.0021 \quad \underline{\underline{Ans}}$$

d.

x	-1	3
y	-2	7.333

% Find x when y=0

	y	x	Δx
y_0	-2	-1 $\rightarrow a_0$	$\frac{3+1}{7.333+2} = 0.429 \rightarrow a_1$
y_1	7.333	3	

$$\therefore x = a_0 + a_1 (y - y_0)$$

$$= -1 + 0.429 (y + 2)$$

$$= -1 + 0.429 (2)$$

$$= -0.142 \quad \underline{\underline{Ans}}$$

Lagrangian Interpolation:

(x)	Age	35	42	50
(y)	Salary	237	349	427

- a. Use Lagrangian polynomial to find salary at age 45.
b. " " " " " age when salary is 300.
c. MATLAB code using "polyfit", "polyval" to find salary at the age of 37, 45, 49.

Soln:

a.

x	35	42	50
y	237	349	427

$$y = \frac{(x-42)(x-50)}{(35-42)(35-50)} (237) + \frac{(x-35)(x-50)}{(42-35)(42-50)} (349) + \frac{(x-35)(x-42)}{(50-35)(50-42)} (427)$$

Put $x = 45$

$$y = \frac{(45-42)(45-50)}{(35-42)(35-50)} (237) + \frac{(45-35)(45-50)}{(42-35)(42-50)} (349) + \frac{(45-35)(45-42)}{(50-35)(50-42)} (427)$$

$$= \boxed{} + \boxed{} + \boxed{}$$

$$= \boxed{384.5}$$

⑥

y	237	349	427
x	35	42	50

$$x = \frac{(y-349)(y-427)}{(237-349)(237-427)}(35) + \frac{(y-237)(y-427)}{(349-237)(349-427)}(42) + \frac{(y-237)(y-349)}{(427-237)(427-349)}(50)$$

put $y=300$

$$x = \frac{(300-349)(300-427)}{(237-349)(237-427)}(35) + \frac{(300-237)(300-427)}{(349-237)(349-427)}(42) + \frac{(300-237)(300-349)}{(427-237)(427-349)}(50)$$

$$= 38.28 \text{ Am.}$$

c. $x = [35, 42, 50];$

$y = [237, 349, 427];$

$p = \text{polyfit}(x, y, 2);$

$x_1 = [37, 45, 49];$

$y_1 = \text{polyval}(p, x_1)$

— o —

Ch. 10
spline

$$y = y_1 + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

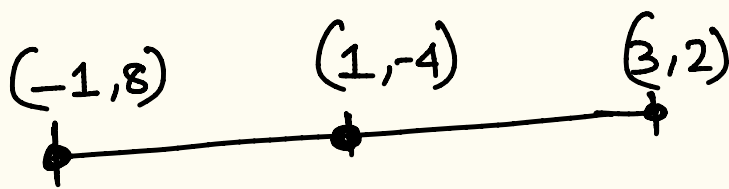
$$y - y_1 = m(x - x_1)$$

Linear spline:

$$(-1, 8), (1, -4), (3, 2)$$

construct linear spline And find $f(1.5)$

soln:



$$f(x) = \begin{cases} 8 + \frac{-4 - 8}{1 - (-1)}(x + 1) & -1 \leq x \leq 1 \\ -4 + \frac{2 - (-4)}{3 - 1}(x - 1) & 1 \leq x \leq 3 \end{cases}$$

$$f(x) = \begin{cases} 8 - 6(x + 1), & -1 \leq x \leq 1 \\ -4 + 3(x - 1), & 1 \leq x \leq 3 \end{cases}$$

$$f(1.5) = -4 + 3(1.5 - 1) = -4 + 1.5 = -2.5 \quad \underline{\text{Ans}}$$

$(-1, -3), (1, 2), (3, 7), (4, 2)$



for $-1 \leq x \leq 1$

$$\begin{array}{cc} (-1, -3) & \& (1, 2) \\ \downarrow & \downarrow & \downarrow & \downarrow \\ x_1 & y_1 & x_2 & y_2 \end{array}$$

$$\therefore y = y_1 + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$= -3 + \frac{2 + 3}{1 + 1} (x + 1)$$

$$= -3 + \frac{5}{2} (x + 1)$$

for $1 \leq x \leq 3$

$$\begin{array}{cc} (1, 2) & \& (3, 7) \\ \downarrow & \downarrow & \downarrow \\ x_1 & y_1 & x_2 & y_2 \end{array}$$

$$y = 2 + \frac{7 - 2}{3 - 1} (x - 1)$$

$$= 2 + \frac{5}{2} (x - 1)$$

for $3 \leq x \leq 4$

$$(3, 7) \& (4, 12)$$

$$y = 7 + \frac{12 - 7}{4 - 3} (x - 3)$$

$$= 7 + 5 (x - 3)$$

$$\therefore f(x) = \begin{cases} -3 + \frac{5}{2} (x + 1) & , -1 \leq x \leq 1 \\ 2 + \frac{5}{2} (x - 1) & , 1 \leq x \leq 3 \\ 7 + 5 (x - 3) & , 3 \leq x \leq 4 \end{cases}$$

$$f(0) = -3 + \frac{5}{2} (0 + 1) = -3 + \frac{5}{2} = -\frac{1}{2} = -0.5 \quad \underline{\underline{Ans}}$$

Cubic Spline

Natural cubic spline $\uparrow$
Clamped " "
Knot-a-not " " $\downarrow$

Natural cubic spline:

$f''(x) = 0$ at the boundary points

f, f', f'' is continuous at interior point.

Q1. Given that

$$f(x) = \begin{cases} A + 3(x-4) + B(x-4)^2 + C(x-4)^3, & 4 \leq x \leq 8 \\ 7 + D(x-8) - 24(x-8)^2 + E(x-8)^3, & 8 \leq x \leq 10. \end{cases}$$

Find A, B, C, D, E

And hence find $f(5)$.

Soln:

$$f(x) = \begin{cases} A + 3(x-4) + B(x-4)^2 + C(x-4)^3, & 4 \leq x \leq 8 \\ 7 + D(x-8) - 24(x-8)^2 + E(x-8)^3, & 8 \leq x \leq 10 \end{cases}$$

$$f'(x) = \begin{cases} 3 + 2B(x-4) + 3C(x-4)^2, & 4 \leq x \leq 8 \\ D - 48(x-8) + 3E(x-8)^2, & 8 \leq x \leq 10 \end{cases}$$

$$f''(x) = \begin{cases} 2B + 6C(x-4), & 4 \leq x \leq 8 \\ -48 + 6E(x-8), & 8 \leq x \leq 10 \end{cases}$$

At the boundary points $f''(x) = 0$ ($x=4, 10$) boundary points

$$f''(4) = 0$$

$$2B + 6C(4-4) = 0$$

$$2B = 0$$

$$\boxed{B = 0}$$

$$f''(10) = 0$$

$$-48 + 6E(10-8) = 0$$

$$12E = 48$$

$$\boxed{\therefore E = 4}$$

$f''(x)$ is continuous at $x=8$

$$\cancel{2B} + 6C(8-4) = -48 + 0$$

$$24C = -48$$

$$\boxed{C = -2}$$

$f'(x)$ is continuous at $x=8$.

$$3 + 0 + 3C(8-4)^2 = D + 0 + 0$$

$$3 + 3(-2)(4)^2 = D$$

$$\boxed{D = 3 - 96 = -93}$$

$f(x)$ is continuous at $x=8$.

$$A + 3(8-4) + 0 + C(8-4)^3 = 7 + 0 + 0 + 0$$

$$\Rightarrow A + 12 + (-2)(4)^3 = 7$$

$$A = 7 - 12 + 128$$

1. $A = \boxed{}$

$$\# f(x) = \begin{cases} 7 + A(x+2) + B(x+2)^2 + C(x+2)^3, & -2 \leq x \leq 3 \\ 5 + D(x-3) - 12(x-3)^2 + E(x-3)^3, & 3 \leq x \leq 5 \end{cases}$$

Hence Find $f(4)$.

Soln:

$$f(x) = \begin{cases} 7 + A(x+2) + B(x+2)^2 + C(x+2)^3, & -2 \leq x \leq 3 \\ 5 + D(x-3) - 12(x-3)^2 + E(x-3)^3, & 3 \leq x \leq 5 \end{cases}$$

$$f'(x) = \begin{cases} A + 2B(x+2) + 3C(x+2)^2, & -2 \leq x \leq 3 \\ D - 24(x-3) + 3E(x-3)^2, & 3 \leq x \leq 5 \end{cases}$$

$$f''(x) = \begin{cases} 2B + 6C(x+2), & -2 \leq x \leq 3 \\ -24 + 6E(x-3), & 3 \leq x \leq 5 \end{cases}$$

at the boundary points: -2, 5

$$f''(-2) = 0$$

$$2B + 6C(-2+2) = 0$$

$$\therefore B = 0$$

$$f''(5) = 0$$

$$-24 + 6E(5-3) = 0$$

$$\Rightarrow 12E = 24$$

$$\therefore E = 2$$

$$\therefore f(x) = \begin{cases} 7 + \frac{98}{5}(x+2) - \frac{4}{5}(x+2)^3, & -2 \leq x \leq 3 \\ 5 - \frac{202}{5}(x-3) - 12(x-3)^2 + 2(x-3)^3, & 3 \leq x \leq 5 \end{cases}$$

$$f(4) = 5 - \frac{202}{5}(1) - 12(1)^2 + 2(1)^3$$

$$= 5 - \frac{202}{5} - 12 + 2$$

$$= \frac{25 - 202 - 60 + 10}{5}$$

$$= \frac{-227}{5} \text{ Ans}$$

$f''(3)$ is continuous:

$$0 + 6C(5) = -24 + 0$$

$$30C = -24$$

$$C = -\frac{4}{5}$$

$f'(3)$ is continuous:

$$A + 0 + 3\left(-\frac{4}{5}\right)(5)^2 = D + 0 + 0$$

$$\Rightarrow A - 60 = D \therefore D = \frac{98}{5} - 60 = -\frac{202}{5}$$

$f(3)$ is continuous:

$$7 + 5A + 0 - \frac{4}{5}(5)^3 = 5$$

$$\Rightarrow 7 + 5A - 100 = 5$$

$$\Rightarrow 5A = 98$$

$$\Rightarrow A = \frac{98}{5}$$

Chapter 6: system of Linear Equations (OBE)

Diagonally dominant:

$$|A_{ii}| \geq \sum_{\substack{i,j=1 \\ i \neq j}}^n |A_{ij}|$$

CO2(4)
↓
PO-b-2

$$4x - 3y - 10z = 20, \quad |10| \geq |4| + |-3| = 7$$

$$x + 4y - 2z = 5, \quad |4| \geq |1| + |-2| = 3$$

$$8x - 3y + z = 15, \quad |8| \geq |-3| + |1| = 4$$

Is the above system diagonally dominant?

$$\left. \begin{array}{l} 8x - 3y + z = 15 \\ x + 4y - 2z = 5 \\ 4x - 3y - 10z = 20 \end{array} \right\} \text{dominant.}$$

Jacobi method:

$$x_{n+1} = \frac{1}{8} (15 + 3y_n - z_n)$$

$$y_{n+1} = \frac{1}{4} (5 - x_n + 2z_n)$$

$$z_{n+1} = -\frac{1}{10} (20 - 4x_n + 3y_n)$$

$$x_0 = 1,$$

$$y_0 = 0.5$$

$$z_0 = 0.2$$

iteration 1 ($n=0$)

$$x_1 = \frac{1}{8} (15 + 3y_0 - z_0) = 2.038$$

$$y_1 = \frac{1}{4} (5 - x_0 + 2z_0) = 1.1$$

$$z_1 = -\frac{1}{10} (20 - 4x_0 + 3y_0) = -1.75$$

iteration 2:

$$x_2 = 2.506$$

$$y_2 = -0.1345$$

$$z_2 = -1.515$$

iteration 3:

$$x_3 = 2.014$$

$$y_3 = -0.134$$

$$z_3 = -0.957$$

clc

clear

$$x(1) = 1; \quad y(1) = 0.5; \quad z(1) = 0.2;$$

for i = 1:3

$$x(i+1) = (1/8) * (15 + 3 * y(i) - z(i));$$

$$y(i+1) = (1/4) * (5 - x(i) + 2 * z(i));$$

$$z(i+1) = (-1/10) * (20 - 4 * x(i) + 3 * y(i)).$$

end

$$sol = [x', y', z']$$

	Candy(x)	ice-cream(y)	chips(z)	
INA	3	2	6	\$ 12.60
MINA	7	1	3	\$ 24.45
TINA	2	4	1	\$ 8.40

$$3x + 2y + \boxed{6z} = 12.60 \quad , \quad |6| \geq |3| + |2|$$

$$\boxed{7x} + y + 3z = 24.45 \quad |7| \geq |1| + |3|$$

$$2x + \boxed{4y} + z = 8.40 \quad , \quad |4| \geq |2| + |1|$$

Gauss-Seidel method:

$$x_{n+1} = \frac{1}{7} (24.45 - y_n - 3z_n)$$

$$y_{n+1} = \frac{1}{4} (8.40 - 2x_{n+1} - z_n)$$

$$z_{n+1} = \frac{1}{6} (12.60 - 3x_{n+1} - 2y_{n+1})$$

$$x_0 = 0, \quad y_0 = 0, \quad z_0 = 0$$

iteration 1:

$$x_1 = 3.493$$

$$y_1 = 0.354$$

$$z_1 = 0.236$$

iteration 2:

$$x_2 = 3.341$$

$$y_2 = 0.351$$

$$z_2 = 0.313$$

itt: 3

$$x_3 = 3.309$$

$$y_3 = 0.367$$

$$z_3 = 0.323$$

— 0 —